Social Mobility Across Generations

A Markov Chain Simulation Study

Portfolio Research Report

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Abstract

This study models **intergenerational social mobility** as a **three-state discrete-time Markov chain** and evaluates the resulting dynamics through Monte Carlo simulation in Excel. **Social position** is represented by **upper**, **middle**, and **lower** levels, and movement between generations follows a **fixed transition matrix**.

The analysis addresses four questions: **whether the long-run social distribution depends on the initial population composition**; **how quickly the influence of an upper- versus lower-level starting position dissipates**; **how long it takes a family line to first cross from one social extreme to the other**; and **what these dynamics imply for decision and policy analysis**. The simulation follows 1,000 family lines over 50 generations and compares three initial distributions.

In the saved simulation realization, generation-50 distributions are closely aligned across starting conditions, with the middle level accounting for approximately 62.7%–64.0% of families. The **Adam–Cooper gap** in the probability of occupying the upper level falls from 46.3 percentage points in generation 1 to approximately one percentage point by later generations. First-passage estimates are **strongly asymmetric:** the mean time from upper to lower is 5.47 generations (95% CI: 5.22–5.72), whereas lower to upper requires 23.55 generations (95% CI: 22.52–24.58). **The results illustrate how aggregate long-run convergence can coexist with substantial short-run persistence and unequal mobility across starting positions.**

Keywords: social mobility; Markov chain; Monte Carlo simulation; transition matrix; first-passage time; convergence; Excel simulation

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1. Background and Research Motivation

Intergenerational mobility concerns the relationship between a family's **social position** in **one generation** and the position attained by its **descendants**. In empirical mobility research, **transition tables** are often used to summarize the probability that children occupy a given socioeconomic category conditional on their parents' category. When the same conditional transition mechanism is extended across multiple generations, the mobility table can be interpreted as the **transition matrix of a discrete-time Markov chain**. This representation is useful because it shifts the analysis from a single parent–child comparison to a dynamic process of persistence, movement, and eventual mixing across generations.

The project studies **a stylized society with three social levels**. **Level 1** is an upper category encompassing executive, managerial, high-administrative, and professional positions; **Level 2** is a middle category including high-grade supervisory, non-manual, and skilled-manual positions; and **Level 3** is a lower category including semi-skilled and unskilled positions. Each person is assumed to have one offspring in the next generation. The model therefore follows **family lines** rather than population growth, allowing the analysis to isolate the consequences of the transition structure itself.

The analytical motivation is twofold. First, the model asks **whether a family's initial social position has a persistent effect or whether that influence eventually dissipates**. Second, it asks **whether mobility is directionally symmetric**. A society can converge to a stable aggregate class distribution while still making movement from the lower to the upper level much more difficult than movement in the opposite direction. Distinguishing these two ideas—aggregate convergence and individual mobility—is central to interpreting the simulation.

Markov-chain representations have a long history in social-mobility research. Contemporary methodological work continues to use transition matrices to characterize intergenerational persistence, equilibrium class distributions, and the rate at which the influence of initial conditions disappears (Song, 2021; Cholli & Durlauf, 2022; Blume et al., 2025). The present project uses the same general logic in a transparent Excel simulation environment.

1.1 Research Questions

Question	Analytical objective
Q1. Long-run distribution	Estimate the generation-50 distribution under three different initial population compositions and assess whether initial conditions matter in the long run.
Q2. Persistence and convergence	Compare the probability that Adam's and Cooper's descendants occupy the upper level in generations 1–10.
Q3. First-passage mobility	Estimate the mean number of generations and 95% confidence interval for upper→lower and lower→upper first passage.
Q4. Implications	Interpret persistence, convergence, and directional asymmetry for managerial, analytical, and public-policy thinking without treating the simulation as causal evidence.

2. Theoretical Framework

2.1 Discrete-Time Markov Chain

Let X_t denote the social level of a family line in generation t, where X_t $\in$ {1, 2, 3}. The model assumes the Markov property: conditional on the current state, the next generation does not depend on earlier family history. Thus,

$$P(X_{t+1} = j \mid X_t = i, X_{t-1}, \ldots, X_0) = P(X_{t+1} = j \mid X_t = i) = p_{ij}.$$

The transition matrix P contains the one-generation conditional probabilities. Every row sums to one, so each row is a complete probability distribution for the offspring's next state conditional on the parent's current state.

Current generation	Next: Upper	Next: Middle	Next: Lower
Upper	45%	48%	7%
Middle	5%	70%	25%
Lower	1%	50%	49%

The diagonal entries describe one-generation persistence. The middle level has the highest probability of remaining in the same state (70%), followed by the lower level (49%) and upper level (45%). The off-diagonal probabilities also show strong directional differences. For example, a lower-level family has only a 1% one-generation probability of moving directly to the upper level, while an upper-level family has a 7% probability of moving directly to the lower level.

2.2 Distribution Dynamics and Long-Run Behavior

If f_t is the row vector describing the distribution of families across the three states in generation t, the population distribution evolves as $f_{t+1} = f_t P$. Repeated multiplication produces $f_1 = f_0 P^t$. For a finite, irreducible, and aperiodic chain, repeated transitions approach a unique stationary distribution that is independent of the initial distribution. The simulation does not assume convergence merely because the theory permits it; instead, it evaluates whether the generation-50 distributions generated from three sharply different starting conditions are empirically close.

The **three initial distributions** are f_0 = [0.5, 0, 0.5], f_0 = [1, 0, 0], and f_0 = [0, 0, 1]. The first represents a population split evenly between the upper and lower levels; the second places every family initially in the upper level; and the third places every family initially in the lower level. Their contrast provides a direct stress test of sensitivity to initial conditions.

2.3 First-Passage Time

Long-run class shares do not fully describe mobility. A family can belong to a system that converges in aggregate yet still require many generations to reach a particular state. First-passage time captures this path-dependent dimension. For a target state j, the first-passage time is the first generation t $\geq$ 1 at which X_t = j. In the workbook, Adam begins in the upper level and the target is the lower level; Cooper begins in the lower level and the target is the upper level. The resulting distributions of first-passage times are summarized by their sample means and 95% confidence intervals.

The workbook uses a 50-generation simulation horizon. If the target state is not reached by generation 50, the calculation engine records 51. This finite-horizon coding is retained in the reported statistics and is therefore an important limitation when interpreting long waiting times.

3. Simulation Design and Excel Implementation

3.1 Inputs and Experimental Design

Element	Specification
States	Upper (1), Middle (2), Lower (3)
Transition matrix	3 × 3 fixed matrix shown in Table 1
Simulation horizon	T = 50 generations
Monte Carlo sample	N = 1,000 family lines
Q1 initial conditions	[0.5, 0, 0.5], [1, 0, 0], [0, 0, 1]
Q2 / Q3 focal families	Adam starts Upper; Cooper starts Lower
Random mechanism	Excel RAND() mapped through cumulative transition probabilities
Uncertainty summary	Sample mean and 95% confidence interval for first-passage time

3.2 Random-State Generation

Excel generates each transition in two steps. First, the probability matrix is converted to a cumulative distribution matrix. Second, RAND() produces a uniform random draw on [0,1), and MATCH() together with INDEX() maps the draw to the appropriate next-state category. Conceptually, the procedure draws X_{t+1} from the categorical distribution represented by the row of P corresponding to X_t. This logic is repeated recursively across generations and independently across the 1,000 simulated family lines.

This implementation is useful for a portfolio setting because the stochastic logic is fully auditable. The transition matrix, cumulative probabilities, simulated paths, generation-level summaries, and first-passage calculations all remain visible in the workbook rather than being hidden inside a software package.

3.3 Output Construction

Q1 counts the generation-50 states and divides by N to estimate the **share of families in each social level**. **Q2** computes, for each generation from 1 through 10**, the proportion of Adam or Cooper simulations occupying the upper level**. **Q3** uses the **first occurrence** of the target state in each family path to construct first-passage times, then reports the sample mean and a normal-approximation 95% confidence interval. **Q4** synthesizes these outputs into an interpretation of persistence, convergence, and mobility barriers.

Because the workbook uses RAND(), recalculation produces a new Monte Carlo realization. The numerical results reported below correspond to the saved simulation realization underlying the finalized portfolio workbook. They should therefore be read as simulation estimates, not immutable constants. The substantive patterns are evaluated through convergence across starting conditions, probability trajectories, and confidence intervals rather than through a single isolated random draw.

4. Results

4.1 Generation-50 Distribution

Initial distribution	Upper	Middle	Lower
[0.5, 0, 0.5]	6.0%	64.0%	30.0%
[1, 0, 0]	6.7%	62.9%	30.4%
[0, 0, 1]	5.5%	62.7%	31.8%

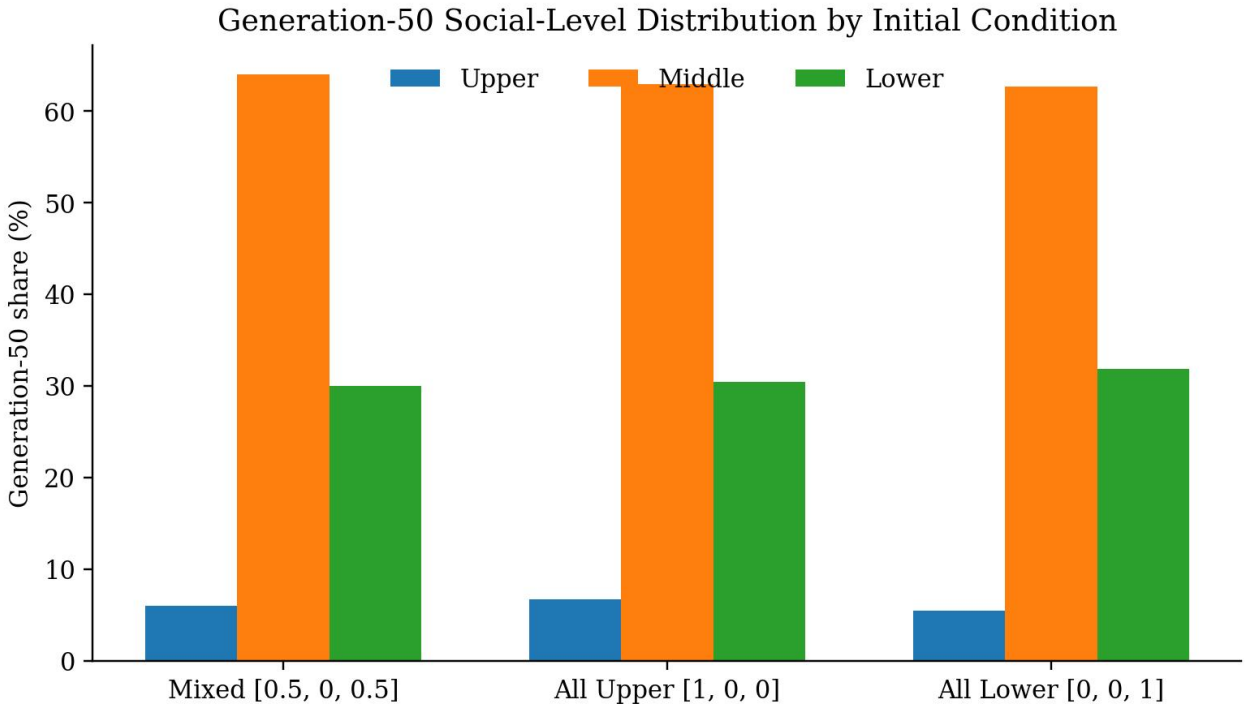


Figure 1. Generation-50 social-level distribution under three initial conditions. Values are taken from the saved Excel simulation realization.

The generation-50 distributions are strikingly similar despite the deliberately extreme differences in starting composition. The upper-level share ranges from 5.5% to 6.7%, the **middle-level** share from **62.7% to 64.0%**, and the lower-level share from 30.0% to 31.8%. The largest spread across the three runs is therefore only 1.8 percentage points. The **initial distribution** strongly determines generation 0 by construction, but its influence on the aggregate class mix is **small** by generation 50.

The **middle level** dominates the long-run simulated distribution. This outcome is consistent with the transition structure: the middle state has a 70% one-generation retention probability, and both the upper and lower states have substantial probabilities of moving into the middle state (48% and 50%, respectively). The transition matrix therefore channels probability mass toward the middle category while allowing continued movement among all three states.

As an analytical validation, the stationary distribution implied directly by the workbook's transition matrix is approximately 6.24% Upper, 62.34% Middle, and 31.42% Lower. The three simulated generation-50

distributions cluster around these values, providing a useful cross-check that the Monte Carlo engine is reproducing the expected long-run behavior of the Markov chain.

4.2 Dissipation of Initial Social Position

Generation	Adam: P(Upper)	Cooper: P(Upper)	Absolute gap
1	47.3%	1.0%	46.3 pp
2	21.4%	3.0%	18.4 pp
3	12.6%	4.6%	8.0 pp
4	9.1%	5.5%	3.6 pp
5	7.1%	5.9%	1.2 pp
6	5.7%	6.5%	0.8 pp
7	6.3%	5.1%	1.2 pp
8	5.1%	6.1%	1.0 pp
9	6.0%	5.4%	0.6 pp
10	6.5%	7.7%	1.2 pp

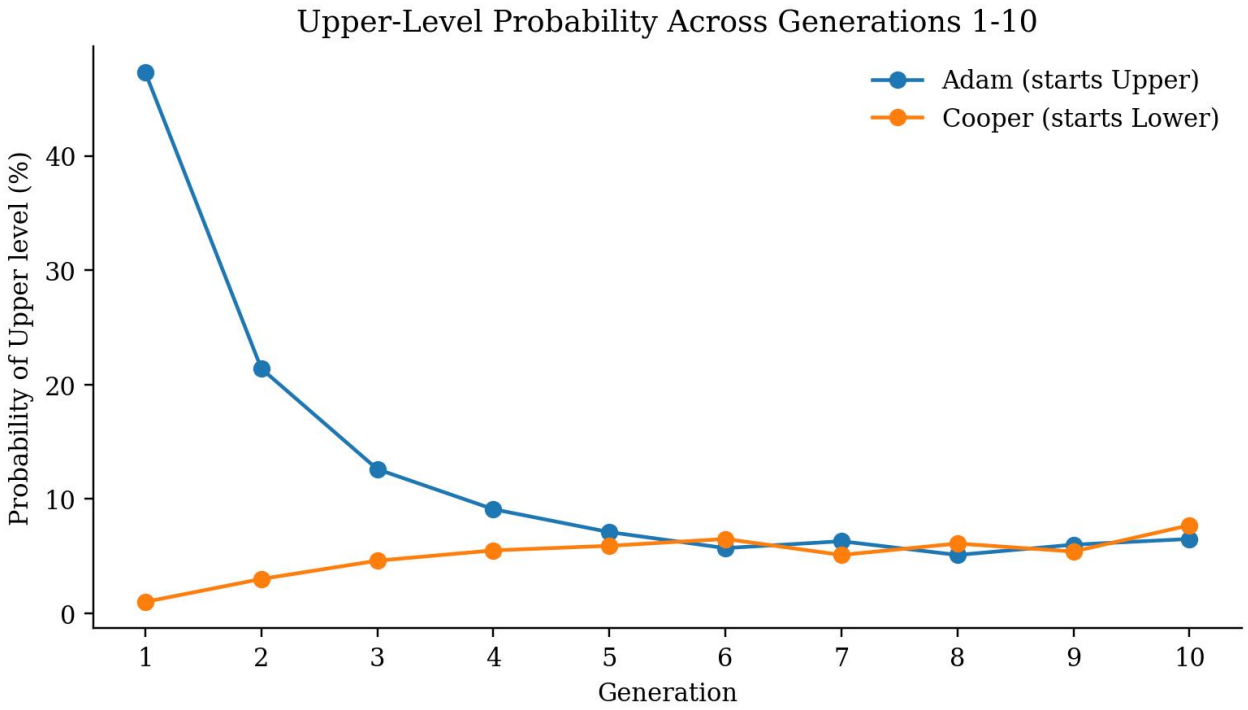


Figure 2. Probability that Adam's and Cooper's descendants occupy the upper level, generations 1–10.

The first generation retains a strong memory of origin. **Adam** starts in the upper level, and the simulated probability that his first-generation offspring remains upper is 47.3%. **Cooper** starts in the lower level, and the corresponding probability of reaching the upper level in one generation is only 1.0%. The generation-1 difference is therefore 46.3 percentage points.

The gap contracts rapidly. By generation 2 it is 18.4 percentage points, by generation 3 it is 8.0 points, and by generation 5 it is only 1.2 points. From generations 6 through 10 the two simulated probabilities fluctuate within a narrow band, with small reversals in ordering. At generation 10, **Adam's** simulated P(Upper) is **6.5%** and **Cooper's** is **7.7%**. The fact that Cooper's point estimate is slightly higher at that generation should not be interpreted as a structural reversal of advantage; with N = 1,000 stochastic paths, late-generation differences of roughly one percentage point are consistent with Monte Carlo variation after the two processes have nearly mixed.

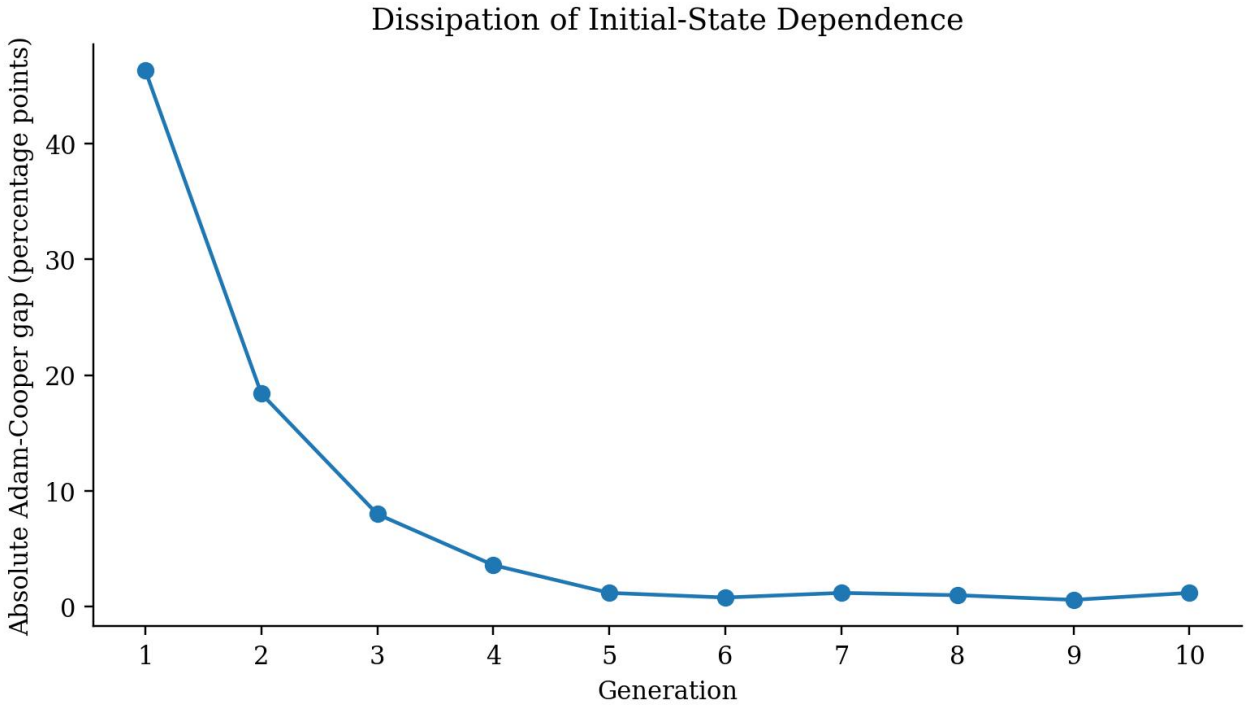


Figure 3. Absolute Adam–Cooper gap in upper-level probability. The rapid decline visualizes the loss of information about the initial state.

This convergence pattern is closely related to the idea of the **Markov process losing memory of its initial condition**. Recent mobility research describes this phenomenon in terms of how rapidly class origins cease to predict later descendants' positions (Blume et al., 2025). In this simulation, **most of the initial-state difference disappears within the first several generations, even though the one-generation transition probabilities themselves remain fixed**.

4.3 First-Passage Mobility Across Social Extremes

Mobility direction	Mean generations	95% CI
Upper → Lower (Adam)	5.47	5.22 – 5.72
Lower → Upper (Cooper)	23.55	22.52 – 24.58

The first-passage analysis reveals a much stronger asymmetry than the long-run class distribution alone. **Adam's** family reaches the **lower level** for the first time after an average of **5.47 generations**, with a 95% confidence interval from 5.22 to 5.72. **Cooper's** family requires an average of **23.55 generations** to reach the **upper level** for the first time, with a 95% confidence interval from 22.52 to 24.58.

The lower-to-upper mean is about 4.3 times the upper-to-lower mean. This difference follows naturally from the transition structure. Direct upper-to-lower movement has a 7% one-generation probability, whereas direct lower-to-upper movement has only a 1% probability. Moreover, movement out of the lower state most often enters the middle state rather than the upper state, so upward passage commonly requires multiple transitions. The first-passage metric therefore captures a practical feature of mobility that a stationary distribution cannot: **even when a destination is reachable in the long run, the expected waiting time can be very different depending on the direction of movement.**

5. Interpretation and Analytical Implications

5.1 Aggregate Convergence Does Not Imply Equal Opportunity

The central analytical finding is that two statements can be true at the same time. **First, the aggregate class distribution converges toward a stable composition that is largely independent of the starting population**. **Second, individual family lines can face highly unequal mobility prospects depending on their initial state**. Long-run convergence is therefore not equivalent to equal opportunity.

This distinction matters whenever a decision maker uses aggregate indicators to evaluate a system. A stable overall distribution can conceal substantial differences in transition speed, path dependence, and waiting time. In this model, the class shares at generation 50 look similar regardless of the starting distribution, yet Cooper's expected wait to first reach the upper level is more than four times Adam's expected wait to first reach the lower level. An analysis based only on the generation-50 distribution would miss this directional barrier.

5.2 Persistence, Memory, and the Middle State

The middle level functions as the system's main mixing state. Its 70% retention probability makes it relatively persistent, while both extremes have roughly one-half probability of sending the next generation to the middle level. As family lines repeatedly pass through the middle state, the distinction between an initially upper and initially lower dynasty weakens. This mechanism explains why the Adam–Cooper probability curves approach one another quickly even though their first-generation probabilities are dramatically different.

From a modeling perspective, this is an example of why multigenerational mobility cannot be summarized by a single upward- or downward-mobility percentage. **The transition matrix jointly determines short-run persistence, long-run composition, and the speed at which origins lose predictive power**.

5.3 Managerial and Policy Relevance

Although the substantive setting is social mobility, the analytical structure has broader managerial relevance. Many business systems can be represented as **movement among discrete states:** customer tiers, credit-risk classes, employee performance categories, supplier quality levels, or stages of organizational health. A **transition matrix** can reveal not only where a system tends to settle, but also **whether movement between desirable and undesirable states is symmetric and how long state changes typically take**.

For public-policy interpretation, **the model suggests that interventions should be evaluated against transition frictions rather than only aggregate class shares**. If lower-to-upper movement is slow because lower-level families have a very small probability of entering the upper state directly, policies aimed at education access, skill development, credit constraints, neighborhood opportunity, or other mobility mechanisms would conceptually seek to alter particular transition probabilities. Taxation and welfare

programs may likewise affect the probability of downward movement or the ability to move from lower to middle and from middle to upper states.

However, the simulation does not estimate the causal effect of any such policy. It contains no policy-treatment variable and assumes the transition matrix is fixed. The appropriate interpretation is therefore diagnostic: **the model identifies where mobility frictions appear in a stylized transition system and shows how changes in those frictions could matter conceptually**. Actual policy evaluation would require empirical data and a design capable of identifying causal effects.

6. Limitations and Extensions

Time-invariant transition probabilities. The model assumes that the same transition matrix applies to every generation. Economic development, education systems, labor-market institutions, demographic change, or major policy reforms could change mobility probabilities over time.

First-order Markov assumption. The next generation depends only on the current generation's social level. Grandparent effects, accumulated wealth, family networks, and persistent unobserved characteristics are excluded.

Three-state simplification. Upper, middle, and lower categories compress substantial within-class variation in income, occupation, education, wealth, and opportunity.

Homogeneous transition process. Every family in a given state is governed by the same transition probabilities. The model does not represent heterogeneity across demographic groups, regions, cohorts, or family characteristics.

Monte Carlo variability. The workbook uses RAND() and N = 1,000 family lines. Recalculation generates a new simulation realization, so reported point estimates vary modestly around the underlying model probabilities.

Finite first-passage horizon. The simulation stops at generation 50 and codes a path that does not reach its target as 51. This creates a practical finite-horizon approximation for very long passage times.

No causal policy identification. Education, taxation, welfare, and related interventions are not explicitly modeled. Policy discussion is interpretive rather than causal.

6.1 Potential Extensions

Several extensions would strengthen the model for research use. First, the simulation could allow transition matrices to vary by cohort or policy regime. Second, family-level covariates could be used to estimate heterogeneous transition probabilities rather than applying one matrix to every family. Third, higher-order Markov specifications could allow grandparents or earlier family history to influence outcomes. Fourth, the simulation could compare counterfactual transition matrices representing alternative interventions and evaluate their effects on stationary distributions, memory curves, and first-passage times. Finally, larger Monte Carlo samples or an explicit random seed in a programmable environment could improve numerical reproducibility.

7. Conclusion

This project demonstrates how a compact **Markov-chain model** can generate several distinct insights about **intergenerational mobility**. Using a **three-state transition matrix**, 1,000 simulated family lines, and a 50-generation horizon, the Excel model shows that **sharply different initial class distributions converge**

toward a similar long-run composition. The **middle level** becomes the dominant state, and the influence of an upper versus lower starting position on P(Upper) declines rapidly across the first several generations.

At the same time, the first-passage analysis shows that **convergence does not eliminate directional inequality**. In the saved simulation realization, movement from the lower to the upper level takes an average of 23.55 generations, compared with 5.47 generations for movement from the upper to the lower level. The project therefore illustrates an important principle of dynamic decision analysis: **a system can appear stable in aggregate while containing substantial asymmetries in the paths experienced by individual units.**

Methodologically, the project also demonstrates a **transparent simulation workflow** in Excel: define states and transition probabilities, convert probabilities to cumulative distributions, generate stochastic paths, summarize long-run and path-dependent outputs, quantify uncertainty, and interpret the results within the limits of the model. This combination of **stochastic modeling, simulation, validation**, and careful **interpretation** makes the project relevant beyond social mobility to a broad class of business and policy problems involving state transitions over time.

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Project workbook: MGT239 Social Mobility Markov-chain simulation, finalized portfolio workbook.